

or mitered turns with a radius at least three times the trace width.

A dedicated ground plane per RF layer is essential to minimise signal noise and to ensure signal integrity. Additionally, trace isolation is crucial; routing digital or power lines near RF traces can lead to mutual inductance and unwanted coupling, resulting in electromagnetic interference.

In essence, high-frequency PCB layout is a balancing act, managing signal integrity, thermal constraints, and electromagnetic compatibility. Every detail, from substrate choice to bend angles, contributes to a wireless product's overall performance and reliability.

Stage 5: Software and firmware

Software design plays a pivotal role in the product development life-cycle, often making the difference between a smooth deployment and a debugging nightmare. The choice of operating system is typically dictated by hardware constraints, with options ranging from real-time operating systems (RTOS) to full-scale Linux or Windows-based platforms.

A modular development approach is strongly recommended, as it enhances code readability and reusability while simplifying future maintenance. Poorly structured legacy code can be notoriously difficult to debug, especially if it is not properly documented.

Version control is non-negotiable in modern development. Tools like Git or SVN should be integrated from day one, ensuring collaborative teams can manage changes efficiently and roll back when needed. Additionally, configuration parameters should be handled systematically to avoid hard-coded values that compromise scalability and flexibility.

Finally, software security and thorough documentation must be prioritised early in the design process. Both are crucial for long-term maintainability, especially in

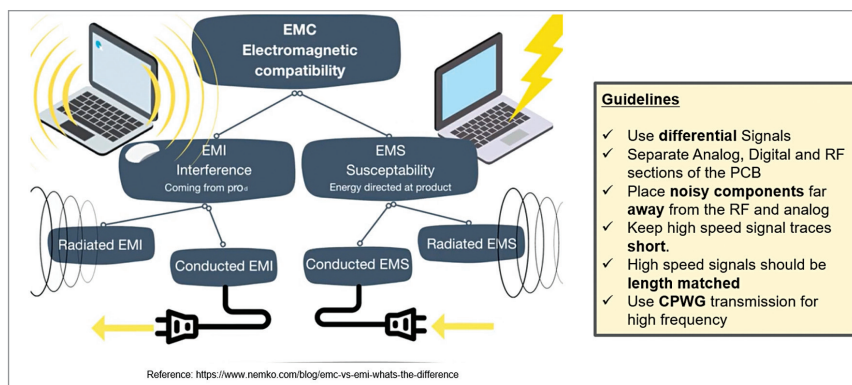


Fig. 6: Safety compliance

connected embedded systems where firmware updates and vulnerability management are ongoing realities.

Stage 6: Safety compliance

Compliance is a critical, often underestimated phase in wireless product development. Regulatory requirements vary by country, but two core standards underpin global certification efforts: EMI (electromagnetic interference) and EMC (electromagnetic compatibility).

EMI testing ensures that devices do not emit unwanted signals that could interfere with other equipment. EMC testing verifies that a device can function reliably in the presence of limited external electromagnetic interference.

Testing is conducted via two main methods: radiated, using antennas to assess emissions over the air; and conducted, where signals are measured through RF cables with controlled attenuation. Both methods determine a product's electromagnetic behaviour and adherence to regulatory thresholds.

To streamline compliance, manufacturers often opt for pre-certified modules or reference-certified designs, particularly when antennas are already characterised. This approach simplifies the certification pathway, reducing cost and time-to-market. In contrast, non-certified solutions require comprehensive testing and formal documentation, which can be complex and resource-intensive.

Certification frameworks differ globally. While FCC (USA) and CE (Europe) dominate Western markets, India follows its own process through the wireless planning and coordination (WPC) wing. For wireless devices operating in unlicensed spectrum, an equipment type approval is mandatory. The WPC process often accepts FCC or CE test reports, though at least one product sample is required for India-specific validation.

In essence, designing for compliance from the outset, especially with EMI/EMC performance and certification strategy in mind, can make or break a wireless product's market viability.

Designing a successful wireless product demands a holistic approach. From balancing hardware, software, and regulatory needs to managing RF considerations, every detail matters. Choosing the right SoC and antenna, ensuring impedance matching, power stability, and EMI/EMC compliance, each decision affects performance and market readiness. Thoughtful component selection, robust layout practices, and early certification planning are not just best practices, they are essential for building reliable, scalable, and commercially viable embedded systems. **EFY**

This article is based on the session titled "Bulletproofing Your Product Design For Wireless Products" delivered by Kousik Barathwaj, Application Specialist from Murata Electronics, at IEW25 in Bengaluru. It has been transcribed and curated by Janarthana Krishna Venkatesan, a journalist at EFY.